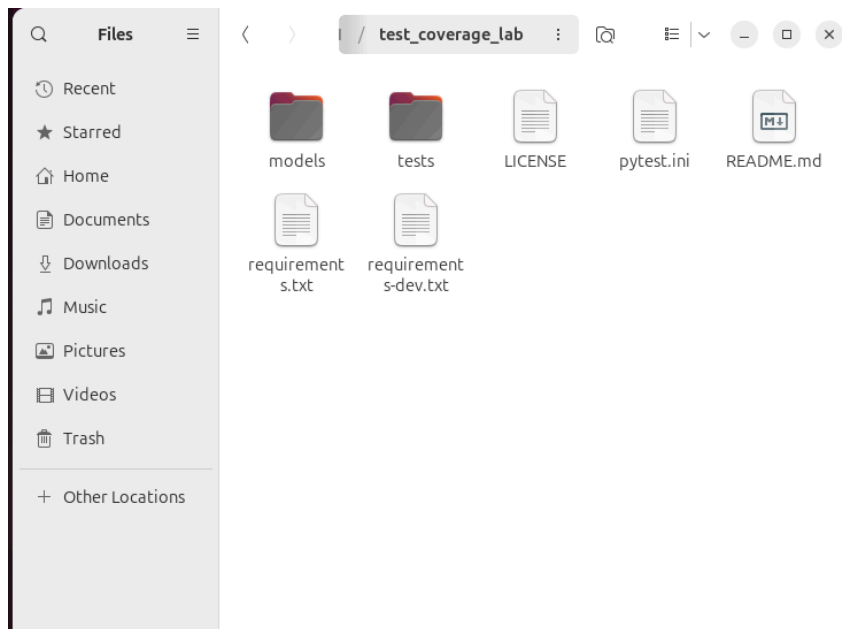


PHASE 1

Set up evidence -



```
(venv) student@student-VirtualBox:~/Group-1/test_coverage_lab$ pytest --cov=models -
-cov-report=term-missing
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/student/Group-1
/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/student/Group-1/test_coverage_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 2 items

tests/test_account.py::test_account_role_assignment PASSED [ 50%]
tests/test_account.py::test_invalid_role_assignment PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                               Stmts   Miss  Cover   Missing
-----
models/__init__.py                   7        0   100%
models/account.py                   58       26    55%   30, 34, 47-49, 53-56, 60-62, 66-68, 72
-76, 80, 84, 94, 98, 102-103
tests/__init__.py                     0        0   100%
tests/test_account.py               35        4    89%   27-30
-----
TOTAL                               100       30    70%

===== 2 passed in 0.45s =====
```

test_coverage_lab PR - <https://github.com/tylervu-unlv/Group-1/pull/37>

test_coverage_lab issue - <https://github.com/tylervu-unlv/Group-1/issues/36>

My PR - <https://github.com/tylervu-unlv/Group-1/pull/55>

My issue/acceptance criteria - <https://github.com/tylervu-unlv/Group-1/issues/38>

Explanation of Test -

The test ensures that the `to_dict()` function actually tests account serialization. It verifies that the expected fields are included in the output and sensitive data is not included.

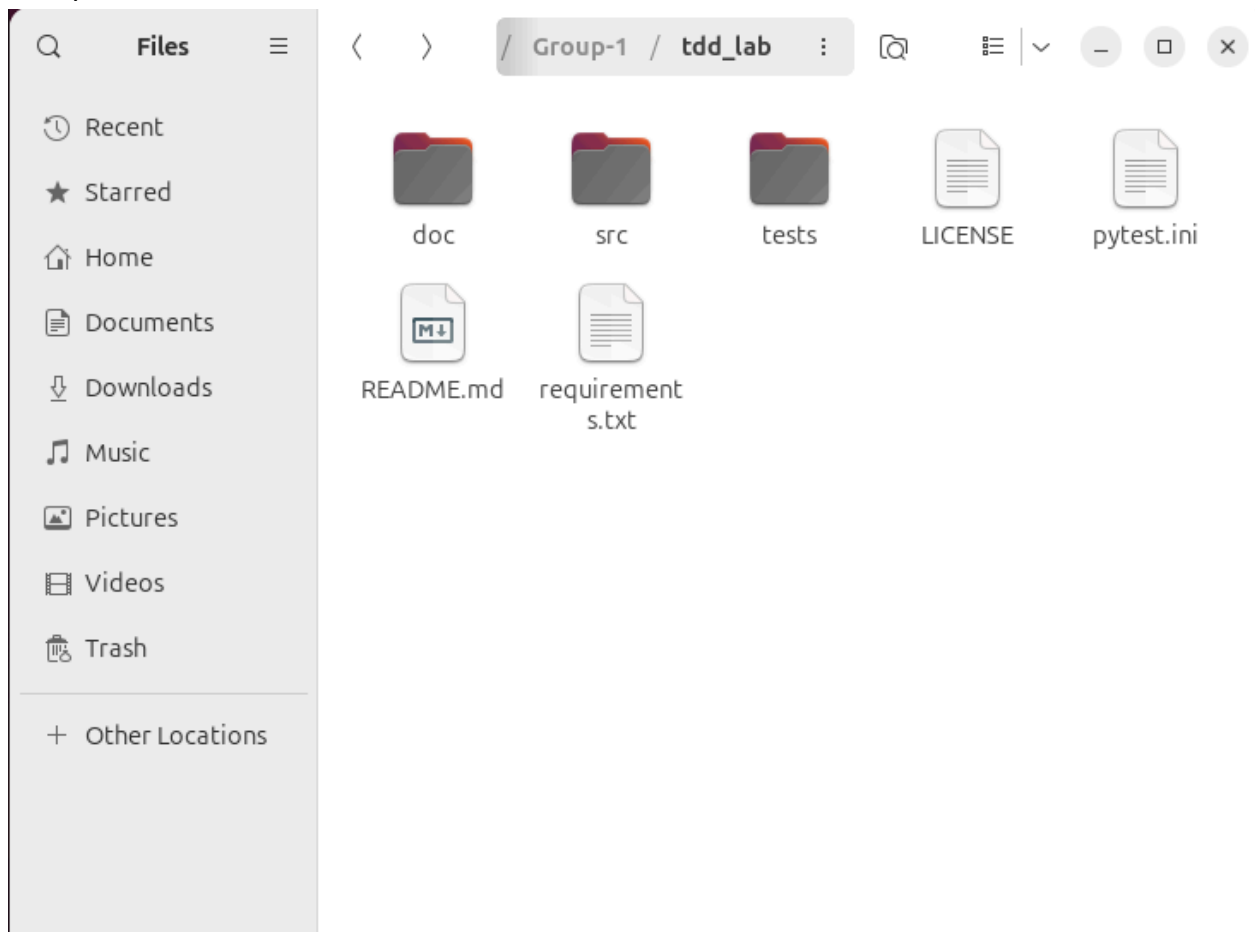
Teammate's PR - <https://github.com/tylervu-unlv/Group-1/pull/50>

PR Review Comment - <https://github.com/tylervu-unlv/Group-1/pull/50#pullrequestreview-5201838074>

Approved PR/Merge Commit - <https://github.com/tylervu-unlv/Group-1/pull/55>

PHASE 2

Set up evidence -



```

(venv) student@student-VirtualBox:~/Group-1/tdd_lab$ pytest --cov=src
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/student/Group-1/test_coverage_lab/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/student/Group-1/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 0 items
/home/student/Group-1/test_coverage_lab/venv/lib/python3.12/site-packages/coverage/control.py:967: CoverageWarning: No data was collected. (no-data-collected); see https://coverage.readthedocs.io/en/7.16.1/messages.html#warning-no-data-collected
  self._warn("No data was collected.", slug="no-data-collected")

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py       3      3     0%    8-11
src/counter.py        2      2     0%    1-3
src/status.py         6      6     0%    2-7
-----
TOTAL                 11     11     0%

===== no tests ran in 0.05s =====

```

Tdd_lab PR - <https://github.com/tylervu-unlv/Group-1/pull/53>

tdd_lab issue - <https://github.com/tylervu-unlv/Group-1/issues/52>

My PR - <https://github.com/tylervu-unlv/Group-1/pull/56>

My issue/acceptance criteria - <https://github.com/tylervu-unlv/Group-1/issues/54>

Explanation of Test -

Test that when a non-alphanumeric counter name is entered, it is rejected. Changes status code (400_BAD_REQUEST) for the correct response. Ensures that the expected fields are included in the output and sensitive data is not included.

RED/GREEN/REFACTOR evidence -

```

(venv) student@student-VirtualBox:~/Group-1/tdd_lab$ pytest --cov=src
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/student/Group-1/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/student/Group-1/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 2 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_create_invalid_counter_name FAILED [100%]

===== FAILURES =====
_____ TestCounterEndpoints.test_create_invalid_counter_name _____

self = <tests.test_counter.TestCounterEndpoints object at 0x75fe9b45a2d0>
client = <FlaskClient <Flask 'src.counter'>>

    def test_create_invalid_counter_name(self, client):
        """It should not create a counter with a non-alphanumeric name"""
        result = client.post('/counters/foo!@#%$')
>       assert result.status_code == status.HTTP_400_BAD_REQUEST
E       assert 201 == 400
E         + where 201 = <WrapperTestResponse streamed [201 CREATED]>.status_code
E         + and    400 = status.HTTP_400_BAD_REQUEST

tests/test_counter.py:39: AssertionError
===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py       3      0   100%
src/counter.py        12      1    92%   19
src/status.py          8      0   100%
-----
TOTAL                 23      1    96%

===== short test summary info =====
FAILED tests/test_counter.py::TestCounterEndpoints::test_create_invalid_counter_name - assert 201 == 400
===== 1 failed, 1 passed in 0.18s =====

```

```

(venv) student@student-VirtualBox:~/Group-1/tdd_lab$ pytest --cov=src
===== test session starts =====
platform linux -- Python 3.12.3, pytest-8.4.2, pluggy-1.6.0 -- /home/student/Group-1/venv/bin/python3
cachedir: .pytest_cache
rootdir: /home/student/Group-1/tdd_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 2 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_create_invalid_counter_name PASSED [100%]

===== tests coverage =====
_____ coverage: platform linux, python 3.12.3-final-0 _____

Name                Stmts   Miss  Cover   Missing
-----
src/__init__.py        3      0   100%
src/counter.py       15      1    93%    19
src/status.py         8      0   100%
-----
TOTAL                 26      1    96%
===== 2 passed in 0.15s =====

```

No refactoring needed because of only one 'if' function addition, which matches the style of the rest of the code.

Reviewer Feedback -

<https://github.com/tylervu-unlv/Group-1/pull/56#pullrequestreview-5253665385>

Teammate's PR - <https://github.com/tylervu-unlv/Group-1/pull/59>

PR Review Comment -

<https://github.com/tylervu-unlv/Group-1/pull/59#pullrequestreview-5240425645>

Evidence examined -

Mainly verified that the RED/GREEN phases were shown and tested. Ensured that proper outputs were shown and that the only affected files were the two scripts. Verified the behavior and tests.

Approved PR/Merge Commit - <https://github.com/tylervu-unlv/Group-1/pull/56>

PHASE 3

Link to branch(repo owner) - <https://github.com/tylervu-unlv/Group-1/tree/tylervu-unlv-ci>

Synchronization Evidence -

```

C:\Users\tyler\Group-1>git pull origin main
From https://github.com/tylervu-unlv/Group-1
 * branch          main          -> FETCH_HEAD
Updating 4d9bdd6..833d078
Fast-forward
 .github/workflows/ci.yml      | 34 +++++
 .gitignore                   | 65 ++++++-
 ci_lab/LICENSE                | 21 +++
 ci_lab/README.md             | 91 ++++++++
 ci_lab/pytest.ini            | 5 +
 ci_lab/requirements-dev.txt   | 4 +
 ci_lab/requirements.txt       | 4 +
 ci_lab/src/__init__.py        | 12 ++
 ci_lab/src/counter.py         | 131 ++++++++
 ci_lab/src/status.py          | 8 +
 ci_lab/tests/__init__.py      | 0
 ci_lab/tests/test_counter.py  | 292 ++++++
 tdd_lab/src/counter.py        | 22 ++-
 tdd_lab/tests/test_counter.py | 56 +++++-
 test_coverage_lab/tests/test_account.py | 72 ++++++
15 files changed, 814 insertions(+), 3 deletions(-)
create mode 100644 .github/workflows/ci.yml
create mode 100644 ci_lab/LICENSE
create mode 100644 ci_lab/README.md
create mode 100644 ci_lab/pytest.ini
create mode 100644 ci_lab/requirements-dev.txt
create mode 100644 ci_lab/requirements.txt
create mode 100644 ci_lab/src/__init__.py
create mode 100644 ci_lab/src/counter.py
create mode 100644 ci_lab/src/status.py
create mode 100644 ci_lab/tests/__init__.py
create mode 100644 ci_lab/tests/test_counter.py

C:\Users\tyler\Group-1>git status
On branch main
Your branch is up to date with 'origin/main'.

```

Set up evidence -

```
C:\Users\tyler\Group-1\ci_lab>python -m pytest --cov=src
===== test session starts =====
platform win32 -- Python 3.14.7, pytest-8.4.2, pluggy-1.6.0 -- C:\Users\tyler\AppData\Local\Python\pythoncore-3.14-64\python.exe
cachedir: .pytest_cache
rootdir: C:\Users\tyler\Group-1\ci_lab
configfile: pytest.ini
plugins: cov-7.1.0
collected 22 items

tests/test_counter.py::TestCounterEndpoints::test_create_counter PASSED [ 4%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_duplicate_counter PASSED [ 9%]
tests/test_counter.py::TestCounterEndpoints::test_retrieve_existing_counter PASSED [ 13%]
tests/test_counter.py::TestCounterEndpoints::test_return_404_for_non_existent_counter PASSED [ 18%]
tests/test_counter.py::TestCounterEndpoints::test_increment_counter PASSED [ 22%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_updating_non_existent_counter PASSED [ 27%]
tests/test_counter.py::TestCounterEndpoints::test_delete_counter PASSED [ 31%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_deleting_non_existent_counter PASSED [ 36%]
tests/test_counter.py::TestCounterEndpoints::test_reset_all_counters PASSED [ 40%]
tests/test_counter.py::TestCounterEndpoints::test_list_all_counters PASSED [ 45%]
tests/test_counter.py::TestCounterEndpoints::test_handle_invalid_http_methods PASSED [ 50%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_counters PASSED [ 54%]
tests/test_counter.py::TestCounterEndpoints::test_top_n_counters PASSED [ 59%]
tests/test_counter.py::TestCounterEndpoints::test_bottom_n_counters PASSED [ 63%]
tests/test_counter.py::TestCounterEndpoints::test_set_counter_to_value PASSED [ 68%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_negative_counter_values PASSED [ 72%]
tests/test_counter.py::TestCounterEndpoints::test_reset_single_counter PASSED [ 77%]
tests/test_counter.py::TestCounterEndpoints::test_prevent_resetting_non_existent_counter PASSED [ 81%]
tests/test_counter.py::TestCounterEndpoints::test_get_total_number_of_counters PASSED [ 86%]
tests/test_counter.py::TestCounterEndpoints::test_counters_greater_than_threshold PASSED [ 90%]
tests/test_counter.py::TestCounterEndpoints::test_counters_less_than_threshold PASSED [ 95%]
tests/test_counter.py::TestCounterEndpoints::test_validate_counter_name PASSED [100%]

===== tests coverage =====
coverage: platform win32, python 3.14.7-final-0

Name                Stmts   Miss  Cover   Missing
-----
src\__init__.py         3      0   100%
src\counter.py        90      7    92%   71, 85, 96, 99-100, 132-136
src\status.py          7      0   100%
TOTAL                  100      7    93%

===== 22 passed in 0.27s =====
```

Issue - <https://github.com/tylervu-unlv/Group-1/issues/76>

My PR - <https://github.com/tylervu-unlv/Group-1/pull/77>

Workflow excerpt -

The screenshot shows a GitHub Actions workflow run for the repository 'tylervu-unlv / Group-1'. The workflow is named 'Add enhancement #1 to ci.yml' and is triggered by a push to the 'main' branch. The run is successful and completed 25 seconds ago. The workflow consists of three jobs: 'build (3.9)', 'build (3.10)', and 'build (3.11)'. The 'build (3.11)' job is currently selected, showing its details. The 'build (3.11)' job is successful and completed 12 seconds ago. The 'build (3.9)' job is also successful and completed 20 seconds ago. The 'build (3.10)' job is successful and completed 13 seconds ago. The workflow is running on a self-hosted runner.

Failing CI log -

<https://github.com/tylervu-unlv/Group-1/actions/runs/35345245127/job/105600170728>

Root-cause summary - <https://github.com/tylervu-unlv/Group-1/actions/runs/35345245127>

Fix commit -

<https://github.com/tylervu-unlv/Group-1/pull/77/changes/8bd97c5906bcef3eca3ff4e6ba1af0f52b924b6f>

Successful CI run -

<https://github.com/tylervu-unlv/Group-1/actions/runs/35345628394/job/105601404081>

Teammates PR - <https://github.com/tylervu-unlv/Group-1/pull/75#pullrequestreview-5252001887>

PR Review Comment - <https://github.com/tylervu-unlv/Group-1/pull/75#pullrequestreview-5252001887>

Evidence examined -

I examined the workflow changes, CI logs, the controlled failure, the tests produced, and the coverage per test.

Reflection -

I didn't know too much about continuous integration, but it was interesting seeing the ins and outs of testing. The GitHub Actions helped me see how one little thing could break the code and helped me debug what exactly needs to be done. The controlled failure task helped me see and understand the CI logs. The whole lab helped explain why tests are so important and why documenting the tests will provide better clarity.